

Is Vitamin D Supplementation Good for Athletes - A Review

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ABSTRACT

The present review states the importance of vitamin D, Role of Vit D in various body functions & supplementation for athletes. Vitamin D has a very crucial role in the muscle metabolism, muscle mending, muscle remodelling & muscle function. The insufficiency of Vitamin D will have a great impact on performance of an athlete where it also has a correlation and risk to develop fractures and cardiac problems. If it is not taken care by the sports scientists & doctor athlete will be at risk.

KEY WORDS: Bone remodeling, Molecular mechanisms, Muscle function, Insufficient Vit D Levels & Vit D receptor.

INTRODUCTION

Vitamin D in the last few years have been considered as a most important part of the modern research & in supplement industry (Owens et al., 2018). The basic function is control of calcium homeostasis; among non-athletic people, we find significant correlation between Vit D & bones (Pilz et al., 2012). There is no specific evidence of research which can establish a link between Vit D & bones, That can be partially attributed to the bone cell stimulus of exercise industry (Owens et al., 2018). The Vit D can influence immune systems response to fight off infections and to recover from harmful exercise by interacting with

extra-skeletal tissues like muscle. Given that many athletes now take supplements, frequently ingesting extremely high amounts of Vit D, it is critical to determine whether too much vitamin D can be harmful to health. According to one theory, hazardous effects only manifest when serum 25[OH]D Levels are higher than 180nmol^{-1} . Excess of this ingestion will lead to intoxication. Finally, we have a contradictory connection between ethnicity, serum Vit D levels, and bone health markers: Indian athletes frequently low amount of serum 25[OH]D levels having no obvious physiological effects (Owens et al.,

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2018). This may be explained by the fact that some ethnic groups have higher quantities of bioavailable (or free) Vit D due to genetic variations in vitamin D binding proteins (Tenforde et al., 2010). Without any pathology, screening might not be essential and could lead to the wrong type of supplementation. In order to define new, cut off points and target concentrations, rechecking of the data must be done to account for bioavailable or "free" Vit D in racial and ethnically varied populations; perhaps we have to consider "set vitamin D free" for the time being.

REVIEW OF LITERATURE

Early in the 20th century, McCollum, a pioneer of nutritional biochemistry, made the discovery of vitamin D. He being a Pioneer to discover the presence of vitamin which is in charge of calcium segregation through his groundbreaking research on experimental rickets. This vitamin was eventually known as vitamin D (Allison, Close, et al., 2015). Vitamin D was previously solely understood to be important for maintaining healthy bones, mostly because to McCollum's research and the rickets epidemic of the same era. Studies that followed established ultraviolet B (UVB) radiation which is the basic source of Vit D demonstrated that only a small amount could be absorbed from the food alone. The development of the Vit D endocrine rapid response to the environments where

humans evolved, which were sun-rich, was thus brought to light (He, Fraser, et al., 2016). Our known information on this system made its importance risen tremendously with the development of new technologies (Girgis et al., 2013). High-throughput gene microarrays and the development of a Vit D receptor knockout mice have revealed a wealth of newly discovered vitamin D targets in a variety of tissues, including bone cells of cardiovascular system & skeletal muscle (Girgis et al., 2013). The proven facts of Vit D are now considered to be far more widespread, despite its historical reputation for playing a classical role in modulating bone turnover. Numerous recently discovered vitamin D actions are relevant to performance of an Athletes, and as a result, Vit D gained attention in the field of sports nutrition (Hassan-Smith et al., 2017). Example, Vit D may help the athlete include enhancing muscular performance and remodelling, Helps in bone health, and reducing the fate of infection (Tenforde et al., 2010). However, it is obvious that Vit D is non-ergogenic aid but rather a biological necessity, hence the usage of exogenous vitamin D in each community depends on the level of vitamin D in that population (Allison, Close, et al., 2015).

Supplementation a new Mark in Sports

The misuse of recommendations & the promotion of fresh, unproven sets of

recommendations yielded a use of supplementary Vit D programmes in top sport (Owens et al., 2018), (Allison, Close, et al., 2015). It makes sense to wonder if supplements are required for all Players. Numerous Research works evaluated 25[OH]D levels of different players around the world for various months of the year. The non-supplemented athlete cohorts differ greatly from one another (Allison, Close, et al., 2015), (Caballero-García et al., 2021). For instance, considerable differences between elite cohorts of rugby players, football players, and jockeys have been seen. The variances may be caused by a variety of elements, including dietary variations, sun exposure, clothes, and lifestyle choices (Allison, Farooq, et al., 2015). Before starting to address the vitamin D deficiency, it's critical to test athletes who are at risk of becoming deficient (Kamycheva et al., 2013). When a requirement for supplemental vitamin D3 has been determined, the proper dosage should be obtained from a source that has undergone the necessary screening (He, Yong, et al., 2016). The recommended dosage for supplements varies greatly and is frequently unclear (Carlberg & Haq, 2018). According to the combined practical experience of the authors, vitamin D dose regimens for elite athletics varies from 1000IU/day - 100000 IU bolus per week (Allison, Close, et al., 2015), (Owens et al., 2018).

Role of Vitamin D

Muscle Mending and Remodelling

The training goal is to deliver a spike which disturbs homeostasis in order to trigger an adaptive response that enhances performance in events (Backx et al., 2017). Therefore, one of the main tenets of a training programme for athletes is to maximise the training stimulus (Cannell et al., 2009). Our group demonstrated in a randomised controlled experiment (RCT) that increasing serum Vit D levels to $[75\text{nmol l}^{-1}]$ with additional Vit D3 at 4000 IU/day had a beneficial impact on force recovery after a session of harmful eccentric exercise (Zarei et al., 2016). Similar findings were seen in investigations that correlated serum Vit D levels with force recovery after strenuous exercise. These findings do not hold back the notion that vitamin D would be significant over a long period of training, but they do suggest that appropriate Vit D exposure can maximise the immediate adaptive response to physically harmful labour (Owens et al., 2018). However, a recent training research offered proof to back up this notion. At a latitude of 56 N, the authors supplemented calcium alone with 800 mg per day or vitamin D3 1920 IU (48lg) in order to increase calcium absorption in 40 untrained young and elderly males (very little sunlight exposure) (Pilz et al., 2012). Molecular mechanisms are necessary to conclusively conclude that Vit D

correlates with muscle which control various elements of muscle remodelling (Owens et al., 2018). The research is still working to identify the important Vit D targets throughout the remodelling process since large studies have concentrated on chemical activities of Vit D in muscle along with research study (translational) (Allison, Close, et al., 2015), (Owens et al., 2018). Global gene expression profiles were examined in one study during mechanical overload-induced skeletal muscle hypertrophy in adult mice (Owens et al., 2018), (Anirudh, 2022). Interestingly, the nuclear receptor-signalling pathway for the vitamin D receptor (VDR) and retinoid X receptor (RXR) significantly increased during the early phases of hypertrophy (Owens et al., 2018). Given the VDR's well-known protein connections are obvious that VDR signalling are involved in the preservation of skeletal muscle mass. In particular, it may be hypothesised that VDR signalling is crucial for satellite cell activity, which is in line with the earlier described in vivo data (He, Fraser, et al., 2016), (Anirudh, 2022). Emerging experimental data are consistent with this idea. First, satellite cells express the VDR, which can control decisions about cell destiny, such as proliferate or divide and preserve the stem cell pool, in satellite cell cultures (Girgis et al., 2013). Vit D deficiency is linked down regulate the notch pathway, a crucial regulator of

satellite cell activation in myogenic cell cultures (Owens et al., 2018). We have demonstrated that 1,25[OH]2D3 enhances the fusion along with movement of human derived skeletal Muscle progenitor cells. Satellite cells exhibit more VDR mRNA and protein expression than mature muscle fibres, indicating that the VDR plays a larger role in muscle progenitors. Undefined signalling axes may be used by the VDR to mediate these effects (Owens et al., 2018). The projected and known VDR-interacting partners (Anirudh, 2022).

Muscle Function

While discussing findings from non-athletic groups, several prior reviews have discussed the facts & functions of Vit D in athletes (Hassan-Smith et al., 2017). Due to their intensive training, athletes often have few rooms for improvement. Because of this, only direct observations of impacts in highly skilled, athletic groups may produce insightful findings pertaining to high performance (Tenforde et al., 2010). A persistent pattern of encouraging results in senior populations may explain claims that Vit D is crucial for muscle function (Owens et al., 2018). According to a meta-analysis, some studies show that adults with Vit D concentrations below 25nmol^{-1} had increased proximal muscular strength (Caballero-García et al., 2021). It's possible that the sarcopenic state of older muscle makes it possible to derive more

quantifiable advantages from maintaining a sufficient vitamin D concentration (Allison, Farooq, et al., 2015), (Anirudh, 2022). According to a different idea, deficiency is increasingly common in the Growing population, which is the sole condition that seriously impairs muscular performance (Kamycheva et al., 2013).

Innate & Acquired Immunity

According to reports, vitamin D is crucial for both innate and acquired immunological functions. Enzyme 1-alpha hydroxylase converts the inactive form of Vit D to its physiologically active form, Vit D, as was previously mentioned. Furthermore, the presence of 1-alpha hydroxylase and the VDR in monocytes, macrophages, neutrophils, T and B lymphocytes implies that Vit D is functional significant to the immune system (He, Yong, et al., 2016). Immune cell activation can be related to levels of 25[OH]D and produced by toll-like receptor cascade activation of pathogenic microbiota Vitamin D specifically increases the gene production of large variety of anti-microbial peptides (AMPs), There are crucial regulators of innate immunity in the immune system (Carlberg & Haq, 2018). In acquired immunity, vitamin D also has an immune modulatory effect on B and T cells. In the innate immune system, AMPs—important proteins that include cathelicidin—help the body fight against

acute illnesses including tuberculosis, influenza, and the common cold (Backx et al., 2017). In order to reduce excessive inflammation and tissue damage, vitamin D is also said to regulate immune-suppressive Th 2 and inflammatory type 1 and type 17 T-helper cells (TH1/TH17) and control the innate immune reaction. Additional research suggests that vitamin D stimulates the oxidative burst in activated macrophages and increases the cytolytic activity of natural killer cells (Cannell et al., 2009). It has been demonstrated that vitamin D3 (100,000 IU) can improve the immune response and limit growth of mycobacteria in vitro. Variations in vitamin D levels could dramatically affect how the immune system reacts (Zarei et al., 2016).

Cardiac Structure & Function

There is some evidence which suggests Vit D insufficiency can raise the risk of cardiometabolic consequences (Pilz et al., 2012). A higher risk of cardiovascular events was linked to lower Vit D concentrations, according to Framingham Offspring Study. With multivariable-adjusted hazard ratios of 1.53 (95% confidence interval (CI) 1.00-2.36) for levels 25 to 37nmol⁻¹ and 1.80 (95% CI 1.05-3.08) for levels 25nmol⁻¹, there was arise in Heart risk across pre-specified thresholds of Vit D deficit. Only participants who had baseline hypertension, nevertheless, were shown to have this connection. Low serum

25[OH]D levels (defined as 37nmol^{-1}) were found to significantly increase the risk of myocardial infarction, according to data from the HPFS (Owens et al., 2018). Scragg et al, Human studies have provided some evidence that a lack of Vit D can be linked to an increased risk of myocardial infarction, as well as indirect relationship Vit D and increase in the normal population, albeit this effect mediated by activity (Allison, Close, et al., 2015), (Anirudh, 2022). While following investigations revealed that blood concentrations of 25[OH]D are significantly linked with LV diastolic dysfunction, the Tromso and Hoorn studies yielded no relation for Vit D and left ventricular (LV) & function. As a result, research has produced conflicting findings about the link between vitamin D and heart function in the general population. Different definitions of vitamin D status, the age of the sample population, definition, and determination may all contribute to the variety of research findings (Hassan-Smith et al., 2017), (Anirudh, 2022). Few studies have looked at the relationship between 25[OH]D concentration and cardiac structure and function in healthy athletes, a population that has frequently been reported to be vitamin D deficient despite some evidence that there is a relationship between vitamin D and cardiac function. In a recent study, our team found that severely 25[OH]D-deficient athletes (those with a 25 nmol^{-1} cut-off) had

smaller aortic root, left atria, intraventricular septum, LV diameter during diastole, LV mass, LV volume during diastole, and right atrial area measurements than insufficient and sufficient athletes (Tenforde et al., 2010). It is still unknown exactly what prevents the 25[OH]D- deficient state from experiencing ventricular hypertrophy. Age of the sample group, vitamin D status (Owens et al., 2018), (Anirudh, 2022).

Fractures & Bones

The metabolically active tissue known as bone is capable of healing structural damage and reacting to mechanical stimuli. Bone remodelling is a step-by-step process which includes resorption, where old bones break down further reversal process start with appearing of mononuclear cells on the surface of the bone further formation of osteoblasts. The most important factor in understanding how vitamin D affects bone is how osteoblast function is regulated. When bones are being remodelled, osteoblasts and osteoclasts reacts with Vit D & PTH which further triggers Phosphate Homeostasis by increasing FGF23 further regulates receptor activator NFB Ligand (RANKL). This entire Process improves the mechanical load by Protein kinase Signalling (Owens et al., 2018). Mechanical loading that athletes experience during practise or competition

is linked to an increase in (BMD) (Caballero-García et al., 2021). Any increase in body mass brought on by exercise promotes bone remodelling and creates mechanically sound bone structure. Athletes bone health is thought to be made up for by the stimulus of loading the musculoskeletal system during high-intensity dynamic sporting activity (Allison, Farooq, et al., 2015). In an ethnically varied athletic population, recent study has found no correlation between blood 25[OH]D concentration and indices of bone health, regardless of the kind of exercise (weight-bearing or non-weight-bearing). This calls into doubt the usefulness of 25[OH]D concentration in predicting bone health in athletes. The 25[OH]D/1,25[OH]2D pathway's genetic polymorphism may be able to explain some of these variations (He, Yong, et al., 2016). Recent studies that showed racial disparities in vitamin D manifestations and indices of bone health lend weight to this idea. Serum 25[OH]D levels that are ideal for maintaining the optimal bone health are still up for dispute. The lowest serum 25[OH]D level that best optimises BMD and/or effectively reduces PTH secretion is commonly used by researchers to establish the threshold for vitamin D sufficiency (Carlberg & Haq, 2018). Observational studies have revealed erratic relationships between BMD and blood 25(OH)D status particularly in populations of people of colour and

athletes (Backx et al., 2017).

MECHANISM

By preventing the synthesis of proinflammatory cytokines like interleukin-6, which encourages the transformation of monocytes into macrophages and consequently heightens the production of other pro-inflammatory cytokines, Vit D can reduce inflammation. As a result, vitamin D lowers the synthesis of cytokines that cause inflammation, such as IFN-, IL-2, and TNF. It is generally known that chronic illnesses and damage to the skeletal muscle are linked to vitamin D deficiency levels (Cannell et al., 2009). Vit D interacts to the intracellular receptor VDR in order to reduce inflammation and numerous important genes are expressed differently by the VDR-vitamin D complex. Through the activity of endothelial NOS (eNOS), the VDR-vitamin D complex also controls NO production (Caballero-García et al., 2021). By encouraging the angiogenic actions of endothelial cells, the controlled generation of NO can enhance endothelial function. Increased Production of arginase -2 competes with eNOS for L-arginine produces ornithine & Urea. In Absence of VDR gene Availability of L-arginine is reduced. However, mounting data appears to show a crucial role for vitamin D in reducing inflammation (Owens et al., 2018). The maintenance of an anti-inflammatory

response depends on a sufficient level of vitamin D in the blood. Regarding this, 1,25(OH)D therapy inhibits the expression of pro-inflammatory cytokines and the VDR-vitamin D complex contributes to the recovery of skeletal muscle function (He, Yong, et al., 2016). The following are the hypothesised mechanisms for vitamin D's ability to increase muscle contraction and restore strength: (a) Directly through binding 1,25(OH)D to VDRs in muscle cells (b) favouring calcium transport in the sarcoplasmic reticulum. However, only rat models have been used to evaluate this last mechanism (Cannell et al., 2009). Endothelial cells have a different role in the process (Zarei et al., 2016). The significance of vitamin D in muscle function is thus shown by the genomic and non-genomic effects of vitamin D on skeletal muscle, accounting for both changes in gene transcription and calcium homeostasis. First of all, 1,25(OH)D is essential for maintaining skeletal muscle homeostasis through the best Ca^{2+} and phosphorus balance. It has been demonstrated that vitamin D plays a critical function in the regulation of skeletal muscle tone in this regard (Caballero-García et al., 2021).

CONCLUSION

A number of authors have claimed Vit D insufficiency is prevailing in athletes. Ogan and Pritchett provided a collection of studies revealing the incidence of

vitamin D insufficiency in this manner (Owens et al., 2018). There is a little information and trails which elucidate the performance improvement with Vit D treatment (Kamycheva et al., 2013). Therefore, when low or insufficient levels of circulating vitamin D (20-30 ng/mL) are restored by supplementation, musculoskeletal benefits result. However, when participants are supplied with vitamin D while having previously normal circulation levels (50 ng/mL), no increases in performance & muscle function is seen (He, Yong, et al., 2016). Otherwise, people with circulation vitamin D levels above 30 have not shown any obvious benefits. Therefore, it is obvious that maintaining optimum vitamin D levels is essential for maintaining both fundamental bodily functioning as well as muscle performance, recuperation, and anti-inflammatory regulation after exercise (Caballero-García et al., 2021), (Anirudh, 2022).

Regarding the ideal supplementation levels, we have few discrepancies. The amount of Vit D needed each day to meet the needs of all the body's tissues and cells has been calculated to be 3000 and 5000 IU (75-125 g/day) (Tenforde et al., 2010). The amounts advised by experts may, however, promote vitamin D storage and boost availability in addition to meeting daily metabolic needs. This dietary approach to lower the pathologies & probably increase athletes performance

(Caballero-García et al., 2021). All of these queries require further investigation. However, Carlberg and Hag suggested that the "Personal Vitamin D Response Index" is connected to supplementation dose (Caballero-García et al., 2021). In our opinion, monitoring the circulation vitamin D levels over an extended period of time is necessary before providing vitamin D supplements

to athletes. The dosage is still a contentious subject (Owens et al., 2018). The Regular doses advised by the Endocrine Society sustained over long period of time interval may be able to replicate the benefits of a short-term mega dose supplementation, based on our experience with athletes (Backx et al., 2017).

REFERENCES

1. **Allison, R. J., Close, G. L., Farooq, A., Riding, N. R., Salah, O., Hamilton, B., & Wilson, M. G. (2015).** Severely vitamin D-deficient athletes present smaller hearts than sufficient athletes. *European Journal of Preventive Cardiology*, 22(4), 535–542.
2. **Allison, R. J., Farooq, A., Hamilton, B., Close, G. L., & Wilson, M. G. (2015).** No association between vitamin D deficiency and markers of bone health in athletes. *Medicine and Science in Sports and Exercise*, 47(4), 782–788.
3. **Anirudh, J. (2022).** Whey as crucial component in rejuvenating athlete health- a review. 10(4), 135–155.
4. **Backx, E., Van Der Avoort, C., Tieland, M., Maase, K., Kies, A., Van Loon, L., De Groot, L., & Mensink, M. (2017).** Seasonal variation in Vitamin D status in elite athletes: A longitudinal study. *International Journal of Sport Nutrition and Exercise Metabolism*, 27(1), 6–10.
5. **Caballero-García, A., Córdova-Martínez, A., Vicente-Salar, N., Roche, E., & Pérez-Valdecantos, D. (2021).** Vitamin D, its role in recovery after muscular damage following exercise. *Nutrients*, 13(7), 1–11.
6. **Cannell, J. J., Hollis, B. W., Sorenson, M. B., Taft, T. N., & Anderson, J. J. B. (2009).** Athletic performance and vitamin D. *Medicine and Science in Sports and Exercise*, 41(5), 1102–1110.
7. **Carlberg, C., & Haq, A. (2018).** The concept of the personal vitamin D response index. *Journal of Steroid Biochemistry and Molecular Biology*, 175, 12–17.
8. **Girgis, C. M., Clifton-Bligh, R. J., Hamrick, M. W., Holick, M. F., & Gunton, J. E. (2013).** The roles of vitamin D in skeletal muscle: Form, function, and metabolism. *Endocrine Reviews*, 34(1), 33–83.
9. **Hassan-Smith, Z. K., Jenkinson, C., Smith, D. J., Hernandez, I., Stuart, A. M.,**

- Crabtree, N. J., Gittoes, N. J., Keevil, B. G., Stewart, P. M., & Hewison, M. (2017). 25-hydroxyvitamin D3 and 1,25-dihydroxyvitamin D3 exert distinct effects on human skeletal muscle function and gene expression. *PLoS ONE*, 12(2), 1–20.
10. He, C. S., Fraser, W. D., Tang, J., Brown, K., Renwick, S., Rudland-Thomas, J., Teah, J., Tanqueray, E., & Gleeson, M. (2016). The effect of 14 weeks of vitamin D3 supplementation on antimicrobial peptides and proteins in athletes. *Journal of Sports Sciences*, 34(1), 67–74.
 11. He, C. S., Yong, X. H. A., Walsh, N. P., & Gleeson, M. (2016). Is there an optimal Vitamin D status for immunity in athletes and military personnel? *Exercise Immunology Review*, 22, 41–62.
 12. Kamycheva, E., Johnsen, S. H., Wilsgaard, T., Jorde, R., & Mathiesen, E. B. (2013). Evaluation of serum 25-hydroxyvitamin d as a predictor of carotid intima-media thickness and carotid total plaque area in nonsmokers: The tromsø study. *International Journal of Endocrinology*, 2013.
 13. Owens, D. J., Allison, R., & Close, G. L. (2018). Vitamin D and the Athlete: Current Perspectives and New Challenges. *Sports Medicine*, 48(s1), 3–16.
 14. Pilz, S., Kienreich, K., Tomaschitz, A., Lerchbaum, E., Meinitzer, A., März, W., Zittermann, A., & Dekker, J. M. (2012). Vitamin D and cardiovascular disease: Update and outlook. *Scandinavian Journal of Clinical and Laboratory Investigation*, 72(SUPPL. 243), 83–91.
 15. Tenforde, A. S., Sayres, L. C., Sainani, K. L., & Fredericson, M. (2010). Evaluating the relationship of calcium and vitamin D in the prevention of stress fracture injuries in the young athlete: A review of the literature. *PM and R*, 2(10), 945–949.
 16. Zarei, A., Morovat, A., Javaid, K., & Brown, C. P. (2016). Vitamin D receptor expression in human bone tissue and dose-dependent activation in resorbing osteoclasts. *Bone Research*, 4(1), 1–10.